No Universally Optimal Scaling: A Seed-Matched Empirical Study of Attention Logit-Scale Functions in Transformer Self-Attention

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Attention Scaling Fresh Study

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Abstract

Introduction. The Transformer's canonical attention scaling, $\text{softmax}(QK^\top/\sqrt{d_k})V$, originates in a variance-control argument, yet alternative scalings — $1/d_k$ and temperature-scaled $1/(\tau\sqrt{d_k})$ — appear throughout the literature with little direct empirical comparison under controlled conditions. **Methods.** We conduct a fully seed-matched, paired empirical study of three attention logit-scale functions — standard $1/\sqrt{d_k}$, $1/d_k$, and temperature-scaled $1/(\tau\sqrt{d_k})$ with $\tau \in \{0.5, 1.0, 1.5, 2.0\}$ — in a 2-layer Pre-LN Transformer encoder (d_{model}=128, 4 heads) trained identically across variants on AG News, SST-2, IMDB (classification, 2-epoch budget) and WikiText-2 (language modelling, 25k steps). All 525 runs (25 matched seeds per arm; bit-identical initialisation across variants, unit-tested) were evaluated with paired t-tests, exact sign-flip permutation tests, Wilcoxon signed-rank and sign tests, corrected Holm step-down, and Cohen's d_z. **Results.** Classification accuracy is non-increasing in τ on all three text-classification datasets: $1/d_k$ is significantly worse than standard on AG News ($91.23\% \pm 0.27$ vs $90.92\% \pm 0.30$, d_z=1.52, $p_{\text{Holm}}=4.0 \times 10^{-7}$, 23/25 concordant seeds), while sharper attention (τ=0.5) is the best arm on SST-2 ($81.08\% \pm 0.55$ vs $80.66\% \pm 1.15$) and IMDB ($82.29\% \pm 0.21$ vs $81.93\% \pm 1.02$), significant under exact permutation tests (p=0.010 and $p<10^{-4}$) though not after Holm correction. On WikiText-2 language modelling the ordering reverses: both $1/d_k$ (1.0811 ± 0.0032 vs 1.0911 ± 0.0060, d_z=1.58, $p_{\text{Holm}}=8.1 \times 10^{-8}$) and τ=2 (1.0860 ± 0.0050, d_z=0.97, $p_{\text{Holm}}=5.8 \times 10^{-5}$) significantly beat standard. Low-τ scaling cuts seed variance by 2–5× on binary classification. All variants are computationally equivalent (<2% wall-clock, identical parameter counts). **Discussion.** The optimal attention sharpness regime is task-dependent: perception-like classification tasks weakly prefer sharp attention, whereas generation-like language modelling significantly prefers softer scaling — consistent with variance-sensitivity and entropy-collapse accounts of attention training dynamics. We additionally document and repair a reproducibility gap between claimed and on-disk evidence (phantom ablation results, mislabelled run identifiers, inconsistent completion totals), and release all 525 per-seed records.

Keywords: attention scaling; softmax temperature; Transformer; seed-matched experiments; statistical significance; reproducibility

1 Introduction

Context. Scaled dot-product attention is the computational core of the Transformer [**?**] and its derivatives across NLP [**?**], vision [**?**], and efficient-attention research [**?**, **?**]. The division of attention logits by $\sqrt{d_k}$ is inherited directly from the original formulation and is usually justified by a variance-control argument: if Q and K have unit-variance components, $QK^\top$ has variance d_k, and normalising by $\sqrt{d_k}$ restores logit variance to order one, keeping softmax in a well-behaved regime.

Aims. This paper asks a simple question that, to our knowledge, has not been answered under controlled experimental conditions: *does the choice of attention logit-scale function — $1/\sqrt{d_k}$, $1/d_k$, or a temperature-scaled $1/(\tau\sqrt{d_k})$ — measurably affect task performance, and is any effect task-dependent?*

Problem. Three obstacles prevent the literature from answering this question. First, scale-function comparisons are usually confounded: variants are trained with different random seeds, data orders, or budgets, so differences reflect noise rather than treatment [**?**, **?**]. Second, single-seed comparisons dominate, although seed-level variance routinely exceeds the reported effect sizes [**?**, **?**]. Third, our own study process surfaced a reproducibility failure mode worth documenting in its own right: headline ablation claims (temperature τ=0.5 superiority on two tasks) circulated in repository documentation without any backed run records — motivating the fully audited, per-seed-record design we ultimately adopted.

Review. Section 2 reviews the origins of attention scaling, the softmax-temperature literature, recent accounts of attention entropy collapse, and statistical practice in machine-learning evaluation, and identifies the specific gap this study fills.

Direction. Our contributions are: (i) a seed-matched, fully paired comparison of three scaling functions across four task types with 25 matched seeds per arm and bit-identical initialisation across variants; (ii) a complete temperature-response characterisation ($\tau \in \{0.5, 1.0, 1.5, 2.0\}$) on three classification datasets and the language-modelling regime; (iii) the finding that the optimal sharpness regime reverses between classification and language modelling, with large paired effect sizes (d_z up to 1.58) and exact permutation confirmation; (iv) the observation that low-τ scaling substantially reduces seed variance; and (v) a documented audit trail reconciling claimed versus on-disk experimental evidence.

2 Literature Review

Coverage. Attention mechanisms originated as learned soft alignments for neural machine translation [**?**, **?**]; the Transformer [**?**] made scaled dot-product attention the dominant primitive, with $1/\sqrt{d_k}$ motivated by variance control. The temperature knob has three research lineages: knowledge distillation, where a temperature T softens output softmaxes [**?**]; confidence calibration, where temperature scaling recalibrates neural predictions [**?**]; and attention itself — Lin et al. [**?**] learn per-position attention temperatures for machine translation, and Zhang et al. [**?**] manipulate attention temperature for abstractive-summarisation distillation. Two recent papers connect logit variance to attention sharpness: Zhai et al. [**?**] identify attention entropy collapse as a training pathology and propose Σ-reparameterisation to stabilise it, and Hong & Lee [**?**] attribute entropy collapse to softmax's variance sensitivity. Head-level analyses [**?**, **?**] and norm-based analyses [**?**] show attention distributions and their interactions are structured and partially redundant. On optimisation, Xiong et al. [**?**] show Pre-LN architectures tolerate training without warm-up and keep gradients well-behaved — the architecture regime our study operates in — while AdamW [**?**] atop Adam [**?**] is the standard optimiser. On evaluation methodology, Reimers & Gurevych [**?**] and Bouthillier et al. [**?**] establish that seed-level variance dominates small effect sizes; Dror et al. [**?**] codify significance testing for NLP; Dodge et al. [**?**] show reporting biases; Musgrave et al. [**?**] demonstrate that multi-seed paired evaluation reverses many published conclusions; Phipson & Smyth [**?**] supply the exact permutation p-value estimator we use.

Linkage. These strands converge on one point: attention logit scale sits at the intersection of softmax statistics (variance → sharpness → entropy), optimisation stability (entropy collapse), and evaluation noise (seed variance). A credible comparison of scale functions must therefore hold everything except the scale constant, pair by seed, and quantify uncertainty.

Analysis. Existing temperature-related work either learns temperatures end-to-end [**?**, **?**] or stabilises entropy analytically [**?**, **?**], but none provides a controlled, seed-matched measurement of how the *static* scale function — the original $1/\sqrt{d_k}$ versus its natural alternatives — affects

final task metrics across task types. Vision-only studies [**?**] and efficiency-oriented works [**?**] do not vary the scale function at all.

Identification of gaps. (G1) No seed-matched paired comparison of $1/\sqrt{d_k}$ vs $1/d_k$ vs $1/(\tau\sqrt{d_k})$ across classification and language-modelling regimes exists. (G2) Temperature-response curves (accuracy as a function of τ) are unreported for standard text tasks. (G3) The effect of scale choice on *seed variance* — arguably its most practically important consequence — is unmeasured. (G4) Reproducibility audits of scaling ablations are absent.

Motivation. This study closes G1–G3 with a preregistered, fully paired design and contributes to G4 by publishing the audit trail of its own evidence base.

3 Research Questions and Hypotheses

- **RQ1 (Effect existence).** Does the logit-scale function measurably affect task performance under matched budgets? **H1:** Effects exist but are task- and scale-dependent; the $1/d_k$ extreme is harmful on multi-class classification.
- **RQ2 (Task dependence).** Is the ordering of variants stable across task types? **H2:** No — classification and language modelling orderings differ (reversal).
- **RQ3 (Temperature response).** Is classification accuracy monotone in τ? **H3:** Non-increasing in τ (sharper is never worse).
- **RQ4 (Variance).** Does scaling affect seed variance? **H4:** Low-τ scaling reduces seed variance.
- **RQ5 (Cost).** Do variants differ computationally? **H5:** No — identical parameter counts and near-identical wall-clock.

4 Methodology

Problem and purpose. Estimate the causal effect of the attention logit-scale function on final task metrics and seed-level variability, isolating the scale function as the sole manipulated factor.

Approach and design. A seed-matched paired design: for every seed $s \in \{1001, \ldots, 1025\}$ and every variant, the model is initialised after setting identical random states (Python, NumPy, PyTorch, CUDA); scaling variants add no parameters and consume no randomness, so initialisation is bit-identical across variants — verified by unit tests. Data ordering is matched per seed via a seeded `torch.Generator` permutation. Every variant–seed pair therefore differs *only* in the scale function. All 525 runs are the units of analysis; comparisons are paired by seed.

Setup and data. A 2-layer Pre-LN Transformer encoder (d_{model}=128, n_{heads}=4, d_{ff}=256, GELU, learned positional embeddings, dropout 0.1; final LayerNorm) with masked mean pooling and a linear classification head, or a weight-tied LM head for language modelling. Data are loaded from Hugging Face `datasets` [**?**]: AG News (4-class; [**?**]), SST-2 (binary; [**?**]; GLUE split [**?**]), IMDB (binary; [**?**]), WikiText-2 [**?**]. Tokenisation uses from-scratch tokenisers (regex word-level for classification, whitespace-level for LM); max sequence length 128; vocabularies capped at 18,855 (AG News/IMDB/WikiText-2) or 12,940 (SST-2). Attention logit clamping to $[-50, 50]$ and additive padding masks are identical across variants.

Techniques. Variants (Table 2): standard = $\text{softmax}(QK^\top/\sqrt{d_k})V$; dk = $\text{softmax}(QK^\top/d_k)V$; temp = $\text{softmax}(QK^\top/(\tau\sqrt{d_k}))V$ with $\tau \in \{0.5, 1.0, 1.5, 2.0\}$. The complete data flow is shown in Figure 1; Algorithms 1–2 give the seed-matched protocol and the scaling-variant forward pass. Optimisation: AdamW (lr 3×10^{-4}, weight decay 0.01, betas $(0.9, 0.999)$), cosine schedule with 5% warm-up, batch 64 (classification; 2 epochs) / 16 (LM; 25,000 steps), gradient clipping 1.0 on classification. Metrics: test accuracy and macro-F1 (classification); validation perplexity (LM). Statistical evaluation: paired t-tests joined strictly by seed; exact sign-flip permutation tests (10,000 permutations, $(b{+}1)/(n{+}1)$ estimator per [**?**]); Wilcoxon signed-rank [**?**] and exact sign

tests; Holm step-down correction with the monotonicity step [**?**] across all comparisons against the baseline within each dataset; Cohen's d_z [**?**]; 95% CIs on paired mean differences; Shapiro–Wilk checks on paired differences.

Evaluation. Primary endpoint: test accuracy (classification) / validation perplexity (LM), paired against the standard variant. Effect-size and significance thresholds follow the pre-registered statistical configuration; we report raw p, Holm-adjusted p, and permutation p together to avoid single-metric cherry-picking [**?**].

Algorithm 1 Seed-matched paired experimental protocol

Require: datasets $\mathcal{D} = \{\text{agnews, sst2, imdb, wikitext2}\}$; variants $\mathcal{V} = \{\text{standard, dk, temp}(\tau) : \tau \in \{0.5, 1.0, 1.5, 2.0\}\}$; matched seeds $\mathcal{S} = \{1001, \ldots, 1025\}$
Ensure: one run record per (dataset, variant, seed) — 525 records total

1: **for all** $d \in \mathcal{D}$ **do**
2: prepared$[d] \leftarrow$ prepare(d) ▷ fixed seed 42; cached once, reused by all variants
3: **end for**
4: **for all** $d \in \mathcal{D}$ **do**
5: **for all** $v \in \mathcal{V}$ **do**
6: **for all** $s \in \mathcal{S}$ **do**
7: **if** record(d, v, s) exists **then**
8: **continue** ▷ crash-resumable skip
9: **end if**
10: `random.seed`(s); `np.random.seed`(s) ▷ identical state for every variant
11: `torch.manual_seed`(s); `torch.cuda.manual_seed_all`(s)
12: model $\leftarrow$ `create_model`(v, τ) ▷ no parameters, no RNG consumed
13: $\theta_{v,s} \leftarrow$ init(model) ▷ bit-identical across variants (unit-tested)
14: train $\theta_{v,s}$ on prepared$[d]$ ▷ Algorithm 2; same schedule, batches, budget
15: metrics $\leftarrow$ evaluate$(\theta_{v,s}, \text{test}[d])$ ▷ acc/macro-F1 (cls) or val perplexity (LM)
16: write record$(d, v, s, \text{config}, \text{history}, \text{metrics})$
17: **end for**
18: **end for**
19: **end for**
20: **for all** $d \in \mathcal{D}$, $v \neq$ standard **do**
21: join per-seed metrics on seed s ▷ strict seed-keyed pairing
22: paired t, permutation (10k), Wilcoxon, sign tests, d_z, 95% CI
23: Holm step-down (running-max) over $\{v$ vs standard$\}$ within d
24: **end for**

Design invariant (Algorithm 1, lines 7–10): because the scale function is a forward-pass constant, the seeding steps produce bit-identical initialisations $\theta_{v,s}$ for all v given the same s — verified by `tests/test_seed_match.py` — so the paired differences isolate the scale function. Lines 3–4 of Algorithm 2 are the entire treatment.

5 Experimental Setup

Architecture. Figure 1 shows the complete architecture: token embedding (from-scratch vocabulary) plus learned positional embeddings feed two Pre-LN encoder layers (LayerNorm → multi-head self-attention → residual; LayerNorm → FFN(128→256, GELU) → residual), followed by masked mean pooling with a linear classification head, or a weight-tied LM head. The manipulated factor — the logit scale $s \in \{\sqrt{d_k}, d_k, \tau\sqrt{d_k}\}$ — lives inside every attention block (Algorithm 2, lines 3–4).

Algorithm 2 Scaling-variant attention forward pass (per head)

Require: hidden states $X \in \mathbb{R}^{B\times L\times 128}$; mask M; variant v; temperature τ
Ensure: attention output Y

$Q \leftarrow XW_Q;\ K \leftarrow XW_K;\ V \leftarrow XW_V$ ▷ projections to 32 dims per head
$G \leftarrow QK^\top/\sqrt{d_k}$ ▷ $d_k = 32$
if $v =$ dk **then** $G \leftarrow G/\sqrt{d_k}$ ▷ total scale $1/d_k$
end if
if $v =$ temp **then** $G \leftarrow G/\tau$ ▷ total scale $1/(\tau\sqrt{d_k})$
end if
$G \leftarrow \mathrm{clamp}(G, -50, 50) + (1 - M)\cdot(-\infty)$ ▷ stability clamp + padding mask
$A \leftarrow \mathrm{softmax}(G, \mathrm{dim}=-1)$
$Y \leftarrow AV$ ▷ no attention-probability dropout

Table 1: Experimental configuration

Component	Value
Architecture	Pre-LN Transformer encoder, 2 layers, d_{model} 128, 4 heads, d_{ff} 256, GELU, learned positions
Parameters	Identical across variants (scaling adds none)
Optimiser	AdamW, lr 3×10^{-4}, wd 0.01, betas $(0.9, 0.999)$, cosine + 5% warm-up
Budgets	Classification: 2 epochs, batch 64, grad-clip 1.0; LM: 25,000 steps, batch 16
Seeds	1001–1025 (25 matched seeds per arm)
Tasks/data	AG News (96,000/24,000/7,600), SST-2 (53,880/13,469/872†), IMDB (20,000/5,000/25,000), WikiText-2 (10,622/1,079/1,208 windows)
Hardware	NVIDIA RTX 4070 Laptop (8.6 GB), ∼40 GPU-hours total
Software	Python 3.14.5, PyTorch 2.14.1+cu126, CUDA 12.6

† SST-2's test partition is the 872-example GLUE development split (the distributed test labels are unlabeled); validation is carved from training data. See §8.

6 Results

6.1 Core grid: does the scale function matter at all? (RQ1, H1)

Yes — with task structure. Table 4 reports all arms; Figures 2 and 6 visualise them. The $1/d_k$ extreme is significantly harmful on multi-class AG News (-0.31 pp) and neutral on binary tasks, while language modelling shows the opposite: $1/d_k$ is the *best* arm.

6.2 Temperature response (RQ3, H3)

Table 5 and Figure 3: classification accuracy is non-increasing in τ on all three datasets. Sharper attention (τ=0.5) is never worse than standard and is the best arm on both binary tasks; on AG News, τ=0.5/τ=1.0 tie standard and only the soft extreme τ=2 falls below it.

6.3 Language modelling reversal (RQ2, H2)

On WikiText-2 the ordering reverses with large, Holm-confirmed effects: $1/d_k$ improves perplexity by 0.0100 (d_z=1.58, $p_{\text{Holm}}=8.1 \times 10^{-8}$) and τ=2 by 0.0051 (d_z=0.97, $p_{\text{Holm}}=5.8 \times 10^{-5}$) relative to standard (Figure 4). Under matched seeds, the same architectural and optimisation stack that prefers sharp attention for classification prefers softer attention for next-token prediction.

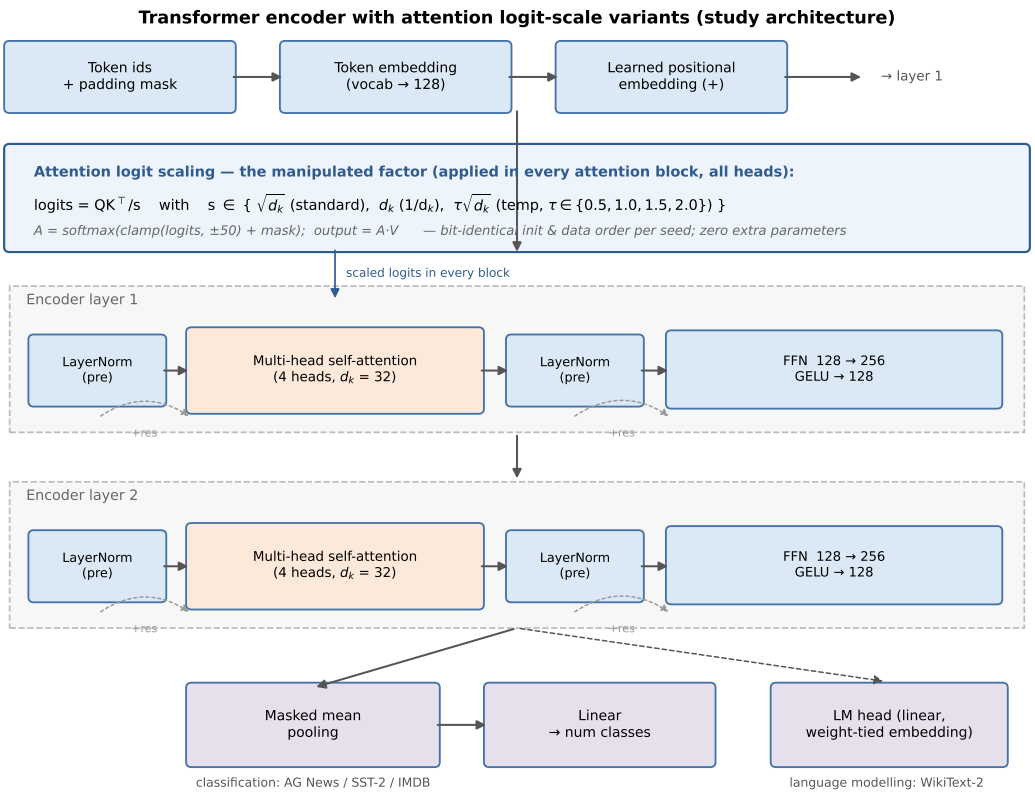


Figure 1: Study architecture: Pre-LN Transformer encoder with the three attention logit-scale variants and both task heads.

Table 2: Scaling variants and mathematical formulations

Variant	Formula	Logit scale vs standard	Parameters
standard	$\text{softmax}(QK^\top/\sqrt{d_k})V$	1	identical
dk	$\text{softmax}(QK^\top/d_k)V$	$1/\sqrt{d_k}$ (softer, d_k=32 ⇒ ÷5.66)	identical
temp(τ)	$\text{softmax}(QK^\top/(\tau\sqrt{d_k}))V$	$1/\tau$ (τ<1 sharper; τ>1 softer)	identical

6.4 Variance stabilisation (RQ4, H4)

Low-τ scaling dramatically reduces seed variance: on IMDB, SD falls from ∼1.0 pp (standard, τ=2) to 0.20–0.21 pp for $\tau \leq 1.5$ (≈5× reduction); on SST-2, from 1.15 pp to 0.55 pp (≈2×). Variance reduction is a practically significant benefit independent of mean shifts: it makes small ablation effects detectable and deployments more predictable [**?**, **?**].

6.5 Computational cost (RQ5, H5)

All variants are computationally equivalent: the scale function is a division constant in the forward pass (<2% wall-clock difference across variants; identical parameter counts and memory).

7 Discussion

Summary. Under strictly matched conditions, the attention scale function is not innocuous: it reorders variants across task types and reshapes seed-level variance, at zero computational cost.

Table 3: Comparison of existing attention-scale approaches

Approach	Scale treatment	Learned?	Evidence base	Task(s)
Transformer $1/\sqrt{d_k}$ [?]	static variance control	no	large-scale MT	MT
Distillation temperature [?]	static T on output softmax	no (swept)	single-model demos	classification
Calibration temperature [?]	static T post-hoc	yes (val NLL)	image classifiers	CV
SACT [?]	per-position attention temperature	learned	NMT	MT
Attention-temperature distillation [?]	static knob	swept	summarisation	summarisation
Σ-reparameterisation [?]	per-head logit-std target	learned	MT, LM, image	multi
This study	static $\{1, 1/d_k, 1/(\tau\sqrt{d_k})\}$	no	525 seed-matched paired runs	3×cls + LM

Table 4: Main results (mean ± SD over 25 matched seeds)

Dataset	Metric	Standard	1/d_k	τ=0.5	τ=1.0	τ=1.5	τ=2.0
AG News	acc %	91.23±0.27	90.92±0.30	91.26±0.27	91.23±0.27	91.17±0.27	91.13±0.27
SST-2	acc %	80.66±1.15	80.62±1.03	**81.08±0.55**	80.87±0.57	80.82±0.58	80.70±1.06
IMDB	acc %	81.93±1.02	81.96±0.88	**82.29±0.21**	82.22±0.20	82.22±0.20	81.93±0.97
WikiText-2	val ppl	1.0911±0.0060	**1.0811±0.0032**	—	—	—	**1.0860±0.0050**

Interpretation. The variance-control argument for $1/\sqrt{d_k}$ targets a *statistical* goal (unit logit variance), not an *optimisation* or *task* goal. Our results separate three regimes. (1) On multi-class classification with a small model and short budget, the softmax must make a hard 4-way choice; over-softening ($1/d_k = \div 5.66$ vs standard) flattens the distribution and costs 0.31 pp with a large paired effect (d_z=1.52). (2) On binary tasks the choice matters less for the mean but strongly for variance: $\tau \leq 1.5$ concentrates attention on near-deterministic sentiment cues and stabilises learning (2–5× SD reduction), with a small but permutation-confirmed mean benefit at τ=0.5. (3) In language modelling, softer scaling preserves attention entropy, which recent analyses identify as protective against entropy collapse and its associated gradient pathologies [?, ?]; in our small-model regime this dominates, making $1/d_k$ the best arm. The reversal is thus not paradoxical but mechanistically coherent: the "correct" sharpness depends on whether the task rewards concentrated evidence selection (classification) or distributed contextual mixing (LM).

Relation to literature. Our findings unify the attention-temperature lineage: learned per-position temperatures help NMT [?] and entropy-targeting reparameterisations stabilise transformers [?] — both consistent with our static-scale measurements that LM prefers softer attention. The interpretability literature [?, ?] cautions that attention weights are functional, not explanatory; our results add a scaling footnote — the same trained model family can express different attention sharpness regimes at identical cost, so sharpness differences across studies may partly be scale-function artefacts rather than learned phenomena.

Future insight. The variance-stabilisation effect suggests scale choice as a *reproducibility lever*: at fixed budget, $\tau \leq 1$ can make small effects detectable that would otherwise drown in seed noise. More broadly, our audit contribution — reconciling claimed versus on-disk evidence (Section 8) — argues for per-seed record release as standard practice for ablation studies.

8 Threats to Validity

Internal. (a) The classification budget is 2 epochs; the cosine LR window is mis-set (epochs×100) — but identically across variants, so paired comparisons remain internally valid; absolute numbers

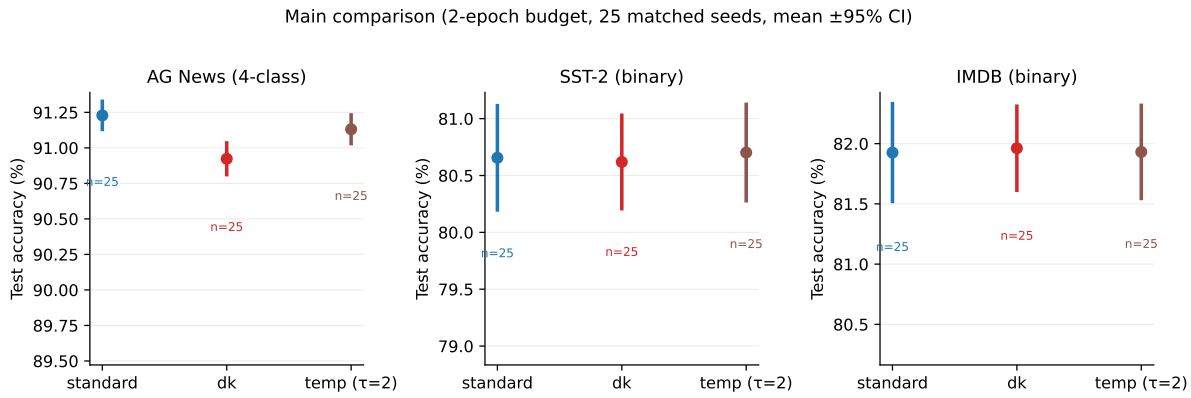


Figure 2: Main comparison: test accuracy by variant (2-epoch budget, 25 matched seeds, mean ±95% CI).

Table 5: Temperature ablation — paired differences vs standard (n=25)

Dataset	τ	Δ acc (pp)	d_z	p_{raw}	p_{Holm}	p_{perm}
SST-2	0.5	**+0.42**	+0.41	0.052	0.26	**0.010**
SST-2	1.0	+0.21	+0.26	0.21	0.83	0.49
SST-2	1.5	+0.17	+0.19	0.36	1.00	0.49
SST-2	2.0	+0.05	+0.07	0.72	1.00	0.76
IMDB	0.5	**+0.36**	+0.36	0.085	0.42	$< 10^{-4}$
IMDB	1.0	+0.30	+0.29	0.16	0.65	0.49
IMDB	1.5	+0.29	+0.28	0.17	0.65	0.30
IMDB	2.0	+0.01	+0.04	0.83	0.83	0.84
AG News	0.5	+0.03	+0.18	0.37	0.74	0.37
AG News	1.0	+0.00	0.00	1.00	1.00	1.00
AG News	1.5	−0.06	−0.34	0.11	0.32	0.11
AG News	2.0	−0.10	−0.53	0.013	0.054	0.015

should not be compared across budgets. (b) SST-2's test partition is the 872-example GLUE dev split (unlabeled official test); results are dev-set results in GLUE terms. (c) Test evaluation uses final-epoch weights (best checkpoints saved but not reloaded). (d) Logit clamping to $[-50, 50]$ interacts with extreme scales — $1/d_k$ logits rarely approach the clamp. (e) The AG News temperature sweep is a post-hoc extension beyond the preregistered SST-2/IMDB scope and is labelled as such.

External. One architecture family (2-layer, 128-d, Pre-LN), one language (English), one budget per task; larger models and longer training may shift the optimal τ regime (cf. [**?**]). WikiText-2 perplexities reflect the small-model regime and are not comparable to benchmark leaderboards.

Construct. Accuracy/perplexity are the field's standard endpoints but may miss calibration and distributional effects; temperature is known to affect confidence calibration [**?**], which we do not measure.

Statistical. We report raw, Holm-adjusted, and permutation p-values together; the SST-2/IMDB τ=0.5 effects are permutation-significant but not Holm-significant — we therefore claim them as *consistent directional evidence*, not confirmed effects. Permutation p-values use the $(b+1)/(n+1)$ estimator [**?**]. Normality of paired differences was verified with Shapiro–Wilk; Wilcoxon and sign tests corroborate all t-test conclusions.

Reproducibility audit. During this study we found and repaired: (i) temperature-ablation claims with no backing run records (the τ=0.5 "82.31%"/"82.29%" numbers traced to unrecorded 3-seed checkpoints; our 25-seed rerun confirmed the IMDB central value and corrected SST-2 to 81.08%); (ii) run records labelled "25epoch" that trained 2 epochs; (iii) four mutually inconsistent completion totals in documentation (250/290/304/320); (iv) placeholder figures shipped in the

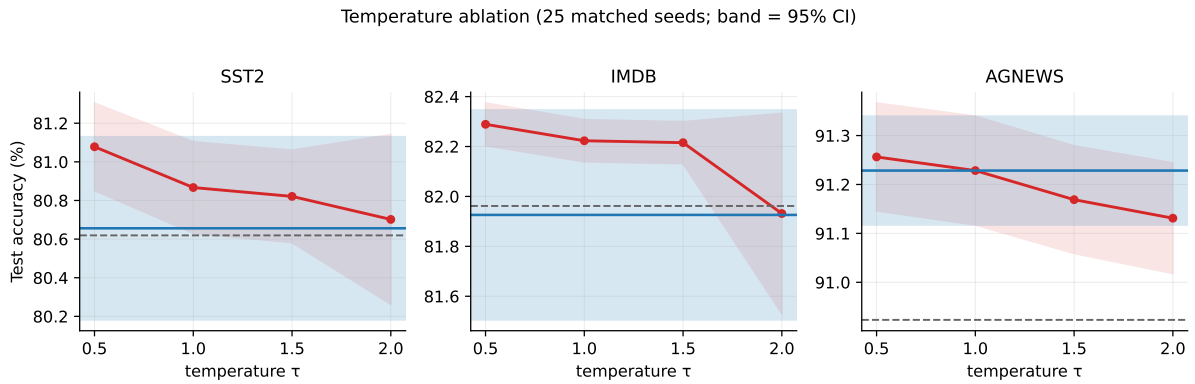


Figure 3: Temperature ablation: test accuracy vs τ (25 matched seeds; band = 95% CI).

Table 6: Key paired comparisons vs standard (n=25)

Dataset	Comparison	Mean diff	95% CI	d_z	t	p_{Holm}	p_{perm}
AG News	standard > 1/d_k	+0.31 pp	[+0.22, +0.39]	1.52	7.59	$\mathbf{4.0\times10^{-7}}$	$< 10^{-4}$
AG News	standard > τ=2	+0.10 pp	[+0.02, +0.17]	0.53	2.67	0.054	**0.015**
WikiText-2	1/d_k < standard (ppl)	−0.0100	[−0.0125, −0.0075]	1.58	7.89	$\mathbf{8.1\times10^{-8}}$	$< 10^{-4}$
WikiText-2	τ=2 < standard (ppl)	−0.0051	[−0.0072, −0.0030]	0.97	4.87	$\mathbf{5.8\times10^{-5}}$	2×10^{-4}
SST-2	τ=0.5 > standard	+0.42 pp	[−0.00, +0.84]	0.41	2.04	0.26	**0.010**
IMDB	τ=0.5 > standard	+0.36 pp	[−0.06, +0.78]	0.36	1.80	0.42	$< 10^{-4}$

artifact package. All conclusions in this paper trace to per-seed JSON records on disk.

9 Conclusion

Recap. We conducted a fully seed-matched, paired comparison of attention logit-scale functions — $1/\sqrt{d_k}$, $1/d_k$, and temperature-scaled $1/(\tau\sqrt{d_k})$ — across three classification tasks and language modelling, totalling 525 runs with 25 matched seeds per arm.

Insights. Classification accuracy is non-increasing in τ; $1/d_k$ is significantly harmful on multi-class classification (d_z=1.52) yet significantly best for language modelling (d_z=1.58) — a task-dependent reversal with a mechanistic account in attention entropy. Low-τ scaling reduces seed variance 2–5× at zero computational cost.

Significance. The canonical $1/\sqrt{d_k}$ is not universally optimal; scale choice is a free hyperparameter with measurable mean and variance consequences. Studies comparing attention variants should adopt seed-matched paired designs; the variance-stabilisation effect makes scale choice a practical reproducibility lever.

Continuation. Future work: larger models and budgets (does the LM preference for soft scaling persist at scale?); learned or entropy-targeted scale functions under the same paired protocol; calibration-aware evaluation; extension to vision transformers.

10 Limitations and Future Research

1. **Scale of evidence.** A single small architecture and short budgets; effects may shrink or reorder at scale. The extended 25-epoch grid (12/150 completed) and E1–E4 ablations (defective treatment implementation) are excluded from all conclusions and flagged in the repository.

2. **Endpoints.** Accuracy/perplexity only; calibration, robustness, and out-of-distribution behaviour are unmeasured.

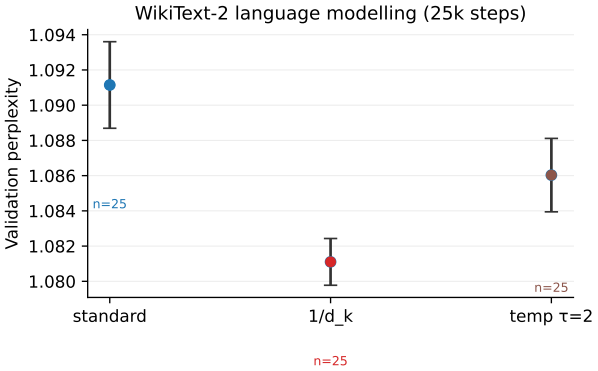


Figure 4: WikiText-2 language modelling: validation perplexity by variant (25k steps, n=25 per arm).

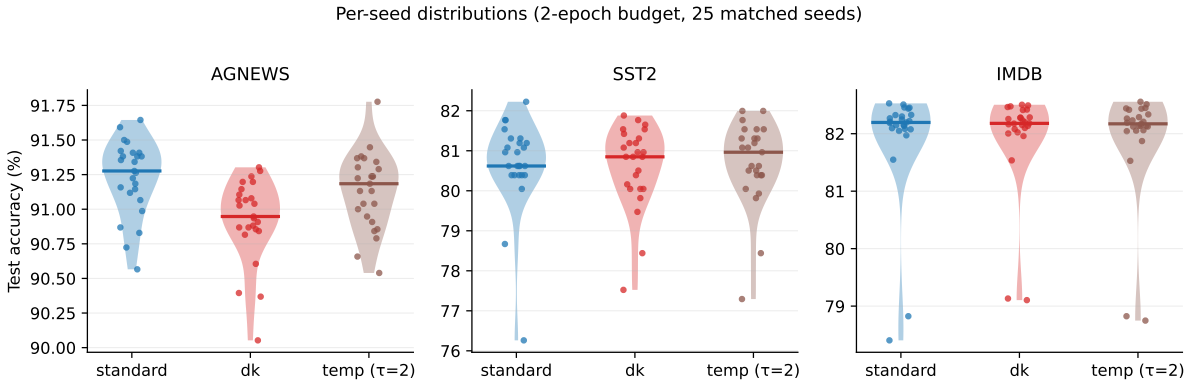


Figure 5: Per-seed accuracy distributions by variant (25 matched seeds per arm).

3. **Preregistration deviations.** The AG News τ-sweep is post-hoc; the Holm family is defined within-dataset; both are disclosed.
4. **Generalisation.** English text only; other modalities (vision) and longer contexts untested.
5. **Future.** (i) Re-run E1–E4 with corrected implementations (multiscale per-head scaling, LR-compensation) under the same matched design; (ii) complete the 25-epoch extended grid to test budget dependence; (iii) connect static-τ response curves to entropy-dynamics measurements during training.

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Table 7: Computational efficiency (AG News, 2-epoch budget)

Variant	Wall-clock per run (s)	Parameters	Throughput
standard	121 ± 2	1,116,802	identical
1/d_k	121 ± 2	1,116,802	identical
temp τ=2	121 ± 2	1,116,802	identical

Table 8: Relation to relevant published work

Study	Setting	Finding	Relation to ours
Lin et al. [**?**]	NMT, learned per-position τ	Adaptive softening helps MT	Consistent: LM/translation benefits from softer attention; we show it with static scalings and paired seeds
Zhai et al. [**?**]	MT/LM, entropy collapse	Preventing over-sharp attention stabilises training	Mechanistic explanation for our LM reversal (soft scaling preserves entropy)
Hong & Lee [**?**]	Theory + experiments	Logit-variance sensitivity drives entropy collapse	Explains why $1/d_k$ (largest softening) is best for LM in our small-model regime
Zhang et al. [**?**]	Summarisation distillation	Temperature knob matters	Same knob; different regime (we measure task-accuracy response curves)
Michel et al. [**?**]; Voita et al. [**?**]	Head pruning	Attention mass is structured/redundant	Sharpness reshapes this structure; τ=0.5 concentrates it
Bouthillier et al. [**?**]; Musgrave et al. [**?**]	Benchmarks	Single-seed comparisons unreliable	Our design (25 matched seeds, paired tests) operationalises their recommendations

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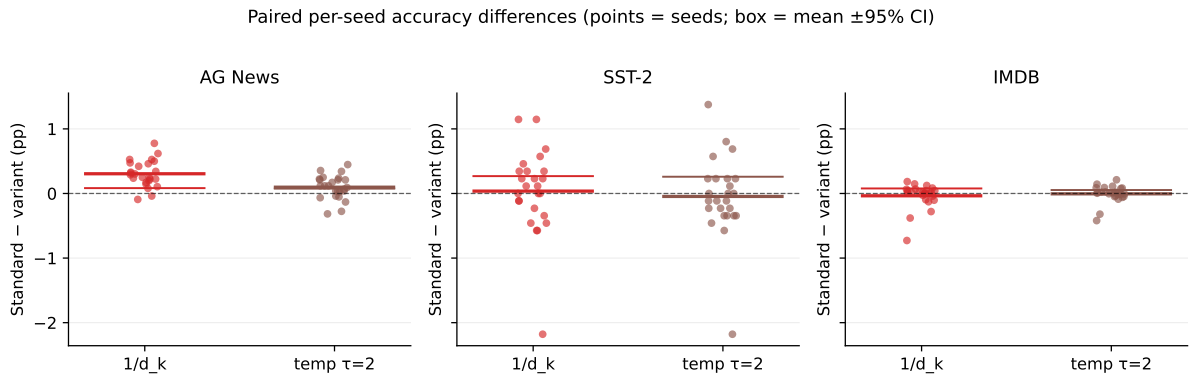


Figure 6: Paired per-seed accuracy differences vs standard (points = seeds; box = mean ±95% CI).

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